

4.3.3 Particle model and pressure

4.3.3.1 Particle motion in gases

Content	Key opportunities for skills development
The molecules of a gas are in constant random motion. The temperature of the gas is related to the average kinetic energy of the molecules. Changing the temperature of a gas, held at constant volume, changes the pressure exerted by the gas.	WS 1.2
Students should be able to: • explain how the motion of the molecules in a gas is related to both its temperature and its pressure • explain qualitatively the relation between the temperature of a gas and its pressure at constant volume.	WS 1.2

4.3.3.2 Pressure in gases (physics only)

Content	Key opportunities for skills development
A gas can be compressed or expanded by pressure changes. The pressure produces a net force at right angles to the wall of the gas container (or any surface).	WS 1.2
Students should be able to use the particle model to explain how increasing the volume in which a gas is contained, at constant temperature, can lead to a decrease in pressure. For a fixed mass of gas held at a constant temperature: $\text{pressure} \times \text{volume} = \text{constant}$ $[p V = \text{constant}]$ pressure, p, in pascals, Pa volume, V, in metres cubed, m^3 Students should be able to calculate the change in the pressure of a gas or the volume of a gas (a fixed mass held at constant temperature) when either the pressure or volume is increased or decreased.	MS 3b, c Students should be able to apply this equation which is given on the Physics equation sheet.

4.3.3.3 Increasing the pressure of a gas (physics only) (HT only)

Content	Key opportunities for skills development
Work is the transfer of energy by a force. Doing work on a gas increases the internal energy of the gas and can cause an increase in the temperature of the gas.	
Students should be able to explain how, in a given situation eg a bicycle pump, doing work on an enclosed gas leads to an increase in the temperature of the gas.	WS 1.2